

# **PIC18F-Based IoT Development Board with Wi-Fi Connectivity and Analog Signal Conditioning**

## **Abstract**

The PIC18F-Based IoT Development Board is a versatile embedded hardware platform designed for sensor interfacing, data acquisition, wireless communication, and industrial monitoring applications. The system integrates a PIC18F microcontroller, ESP8266 Wi-Fi module, LM358-based analog signal conditioning circuits, LCD display interface, digital I/O expansion, and multiple communication protocols such as UART, SPI, and I<sup>2</sup>C. The board is powered through a 24V industrial input and generates regulated 5V and 3.3V supplies for onboard components. The platform enables real-time monitoring, local display, and wireless transmission of sensor data, making it suitable for IoT, automation, and educational applications.

## **1. Introduction**

Embedded systems play a crucial role in modern industrial automation, smart monitoring, and Internet of Things (IoT) applications. Conventional development boards often require multiple external modules for wireless communication, sensor conditioning, and display interfacing. This project addresses these limitations by integrating all essential components into a single PCB platform.

The developed board combines analog signal acquisition, wireless connectivity, user interface components, and industrial-grade power management. It serves as a complete solution for developing IoT-based monitoring and control systems.

## **2. Objectives**

- Design a compact PIC18F-based development board.
- Integrate ESP8266 Wi-Fi connectivity.
- Provide analog signal conditioning for sensor interfacing.
- Support multiple communication protocols.
- Enable real-time display of system information.
- Facilitate industrial monitoring and automation applications.
- Develop a reusable platform for embedded system projects.

## **3. System Components**

### **3.1 PIC18F Microcontroller**

The PIC18F microcontroller acts as the central processing unit. It performs:

- Sensor data acquisition
- Analog-to-digital conversion
- Communication management
- Display control
- Alarm generation
- System monitoring

### **3.2 ESP8266 Wi-Fi Module**

The ESP-01 module provides wireless communication capabilities. It enables:

- Remote monitoring
- Cloud connectivity
- IoT integration
- Wireless data transmission

### **3.3 LM358 Signal Conditioning Circuit**

LM358 operational amplifiers are used to amplify and condition low-level sensor signals before ADC conversion.

Benefits include:

- Improved measurement accuracy
- Noise reduction
- Enhanced sensitivity

### **3.4 LCD Display**

The display provides:

- Real-time sensor readings
- System status information
- Network connection status
- Alert messages

### **3.5 Power Supply Section**

The board accepts a 24V DC industrial input and generates:

- 5V using MC7805 regulator
- 3.3V using AMS1117 regulator

Protection components include:

- Fuse
- Varistor
- Filtering inductor

## **4. Hardware Design**

The PCB consists of the following functional blocks:

### **Power Management Block**

Responsible for voltage regulation and protection.

#### **Microcontroller Block**

Controls all processing and communication activities.

#### **Analog Interface Block**

Interfaces external sensors through LM358 amplifiers.

#### **Communication Block**

Supports:

- UART
- SPI
- I<sup>2</sup>C

#### **Wireless Communication Block**

Provides Wi-Fi connectivity through ESP8266.

#### **User Interface Block**

Includes LCD display, LEDs, and buzzer.

### **5. Working Principle**

1. The board receives a 24V DC supply.
2. Voltage regulators generate 5V and 3.3V rails.
3. The PIC18F initializes all peripherals.
4. Sensor signals enter through analog input channels.
5. LM358 amplifiers condition the sensor signals.
6. ADC converts analog values into digital data.
7. Processed data is displayed on the LCD.
8. Data is transmitted through the ESP8266 Wi-Fi module.
9. LEDs and buzzer indicate system status and alarms.

### **6. Features**

- PIC18F microcontroller-based architecture
- ESP8266 Wi-Fi support
- Analog signal conditioning
- LCD interface
- UART, SPI, and I<sup>2</sup>C communication
- Digital I/O expansion
- 24V industrial power input
- Visual and audible status indicators
- Compact PCB design

### **7. Applications**

#### **Industrial Monitoring**

Monitoring of temperature, pressure, voltage, and current.

#### **Smart Agriculture**

Monitoring soil moisture, water levels, and environmental parameters.

#### **Environmental Monitoring**

Collection of air quality, humidity, and light intensity data.

#### **IoT Systems**

Remote monitoring and cloud-based data logging.

#### **Educational Platforms**

Learning embedded systems, PCB design, and IoT development.

### **8. Advantages**

- Integrated Wi-Fi communication
- Industrial-grade power support
- Multiple communication interfaces
- Improved sensor accuracy through signal conditioning
- Compact and scalable design
- Suitable for various applications

### **9. Limitations**

- Limited processing capability compared to ARM-based systems
- Dependent on ESP8266 for wireless connectivity
- No onboard battery backup
- Linear regulators reduce power efficiency

### **10. Future Enhancements**

- MQTT protocol implementation
- Cloud dashboard integration
- Mobile application support
- Over-the-Air (OTA) firmware updates

- Additional sensor interfaces
- Data logging using SD card

## **11. Pin Mapping**

### **Analog Inputs**

#### **Signal MCU Pin**

AD0 AN0  
AD1 AN1  
AD2 AN2  
AD3 AN3  
AD4 AN4  
AD5 AN5

### **Digital I/O**

#### **Signal MCU Pin**

D0 RB0  
D1 RB1  
D2 RB2  
D3 RB3  
D4 RB4

### **Communication Header**

#### **Signal Function**

TXD UART TX  
RXD UART RX  
SPI1 SPI Data  
SPI2 SPI Clock  
SPI3 SPI Select  
+5V Power  
GND Ground

## **11. Results**

The designed PCB successfully integrates sensor acquisition, display control, wireless communication, and industrial power management into a single platform. The board demonstrates stable operation, reliable sensor interfacing, and effective wireless data transmission, making it suitable for IoT and automation applications.

## **12. Conclusion**

The PIC18F-Based IoT Development Board provides a comprehensive hardware solution for embedded and IoT applications. By integrating analog signal conditioning, wireless communication, display interfaces, and industrial-grade power support, the system offers a flexible platform for monitoring and control applications. The design reduces hardware complexity while improving scalability and ease of development, making it suitable for educational, research, and industrial environments.

## **References**

1. PIC18F Microcontroller Datasheet.
2. ESP8266 ESP-01 Technical Reference.
3. LM358 Operational Amplifier Datasheet.
4. AMS1117 Voltage Regulator Datasheet.
5. MC7805 Voltage Regulator Datasheet.
6. Embedded Systems Design Principles.
7. Internet of Things (IoT) Communication Standards.